

B.Sc. 3rd Semester (Honours) Examination, 2023-24

COMPUTER SCIENCE

Course ID : 31512

Course Code : SH-CSC-302/C-6

Course Title : Operating Systems

[Syllabus 2017-18 & 2022-23]

Time : 1 Hour 15 Minutes

Full Marks : 25

The figures in the right margin indicate marks.

Candidates are required to answer in their own words as far as practicable.

UNIT-I

1. Answer *any five* from the following questions:

5 × 1 = 5

- a) What do you mean by multiprogramming operating system?
- b) Distinguish between the functioning of shell and kernel of an operating system.
- c) Why does an operating system need to make system calls?
- d) Relate thread and process.
- e) Name two methods used for inter-process communication.

[Turn Over]

- f) Why do we need mapping between physical address space and virtual address space? 3.
- g) Name any two types of file allocation methods in operating system.
- h) Name any one method for authentication in operating system.

UNIT-II

2. Answer *any two* from the following questions:

2×5=10

- a) What is semaphore? Compare and contrast shared memory and message passing in the context of inter-process communication.

2+3=5

- b) Draw process state/transition diagram. State the role of interrupt in the context of process state/transition.

4+1=5

- c) Describe segmentation method for managing virtual memory in brief.

- d) Write a short note on protection and security management in operating system.

UNIT-III

3. Answer *any one* from the following questions:

$$1 \times 10 = 10$$

- a) Name 5 major services provided by an operating system and describe any three in brief.

$$1 + 9 = 10$$

- b) Using (i) Shortest Job First and (ii) Round-Robin (quanta = 1 ms) technique determine the average waiting time and turn-around time of three processes whose attributes are given below:

$$3 + 7 = 10$$

Process	CPU Burst Time (ms)	Arrival Time (ms)
P ₁	10	0
P ₂	1	2
P ₃	5	5

BANKURA SAMMILANI COLLEGE

Dept. of Computer Science

INTERNAL ASSESSMENT EXAMINATION – 2023

SEM- 3 (H)

PAPER: Operating System

Full marks: 5

Time: 30 min

Answer any five questions:

1. What do you mean by PCB?
2. Define RAG?
3. What is starvation?
4. Write down the basic principle of RR scheduling algorithm.
5. Write the major goals of operating System?
6. What do you mean by fragmentation?
7. Define starvation. Write down the major Advantages of multi programming.
8. What is response time?

B.Sc. 3rd Semester (Honours) Examination, 2023-24

COMPUTER SCIENCE

Course ID : 31522

Course Code : SH-CSC-302/C-6

Course Title : Operating Systems (Practical)

[Syllabus 2017-18 & 2022-23]

Time : 2 Hours

Full Marks : 15

The figures in the right margin indicate marks.

Candidates are required to answer in their own words as far as practicable.

[Problem = 10, LNB + Viva voce = 05]

1. Perform any **one** from the following experiments:

10×1=10

- a) Write a C-program to implement the best fit memory allocation technique
 - b) Write a C-program to calculate the average waiting time in SJF CPU scheduling
 - ✓ c) Write a C-program to calculate the average turnaround time in priority-based CPU scheduling
 - d) Write a C-program to calculate the average waiting time in round-robin CPU scheduling
 - e) Write an UNIX program using fork() command involving parent and child process which are executing the same program with different codes
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